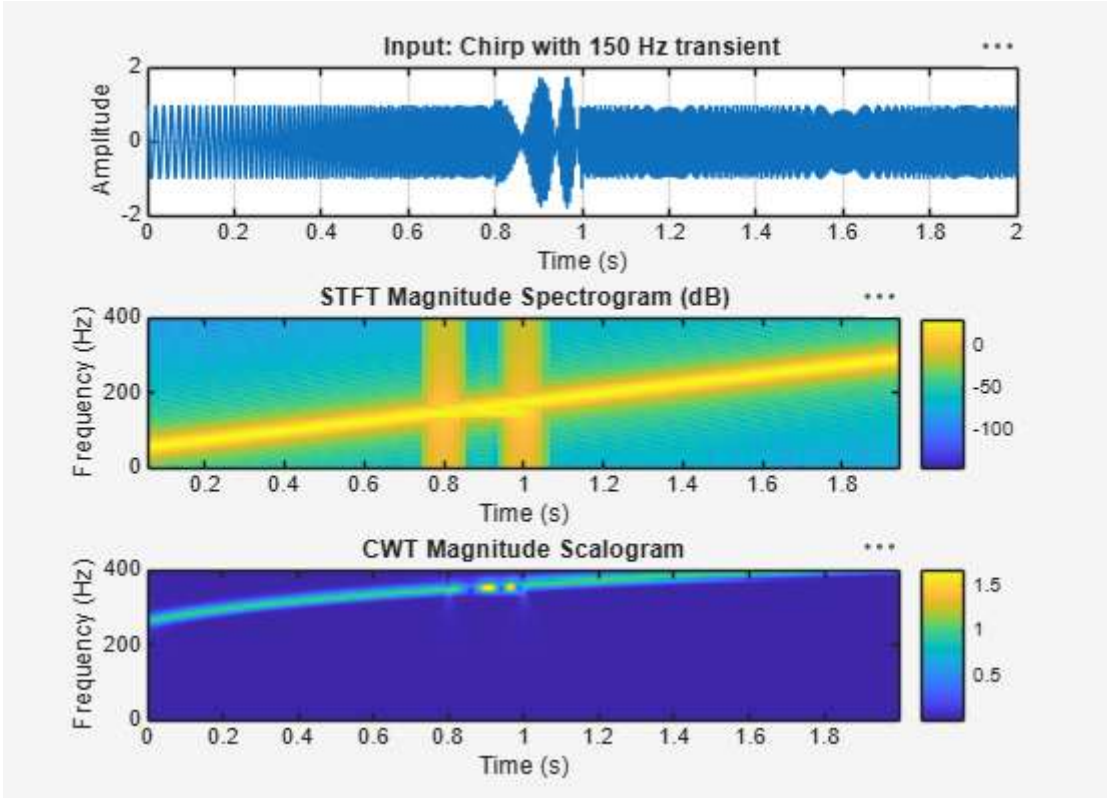
Experiment 20
STFT and Wavelet -Based Time - Frequency Analysis

```
clc; clear; close all;
Fs = 1000;
t = 0:1/Fs:2-1/Fs;
% Non-stationary input: linear chirp from 50 Hz to 300 Hz plus a transient
x = chirp(t,50,t(end),300,'linear');
burst = zeros(size(t));
idx = (t >= 0.8) & (t <= 1.0);
burst(idx) = 0.8*sin(2*pi*150*t(idx));
x = x + burst;
% STFT / spectrogram
win = hann(128,'periodic');
noverlap = 112;
nfft = 512;
[S,F,Tstft] = spectrogram(x,win,noverlap,nfft,Fs);
figure('Name','Experiment 20: STFT and wavelet analysis');
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Input: Chirp with 150 Hz transient');
subplot(3,1,2);
imagesc(Tstft,F,20*log10(abs(S)+eps)); axis xy;
ylim([0 400]); colorbar;
xlabel('Time (s)'); ylabel('Frequency (Hz)');
title('STFT Magnitude Spectrogram (dB)');
% Continuous wavelet transform; requires Wavelet Toolbox
subplot(3,1,3);
[wt,fCWT] = cwt(x,Fs);
imagesc(t,fCWT,abs(wt)); axis xy;
ylim([0 400]); colorbar;
xlabel('Time (s)'); ylabel('Frequency (Hz)');
title('CWT Magnitude Scalogram');
% Report settings used for reproducibility
fprintf('Sampling frequency = %d Hz\n',Fs);
fprintf('STFT window = %d samples; overlap = %d samples; FFT length = %d\n', ...
    length(win),noverlap,nfft);
fprintf('Input contains a chirp from 50 Hz to 300 Hz and a 150 Hz burst from 0.8 to 1.0s.\n');
```

Code



Output